

# **Gravity and Gravity Gradient 3-D Migration v2.0 with Topography**

## **Short Manual**

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### **1. Introduction**

Gravity and gravity gradient migration is a fast tool to get the underground density 3-D distribution from gravity and/or gravity gradient data.

The code uses the migration technique to image the 3D density distribution.

Field gravity and/or gravity gradiometry data can include topography data.

The code is written in Matlab, and it needs to be run in a MATLAB environment.

### **2. What's New?**

The v2.0 version (2014) added an option for doing joint migration of any gravity and gravity gradient components.

The input file added a joint migration indicator and joint migration components description.

### **3. Gravity and/or gravity gradiometry**

Gravitation, or gravity, is a natural phenomenon by which physical bodies attract with a force proportional to their mass. Gravitation is most familiar as the agent that gives weight to objects with mass and causes them to fall to the ground when dropped. Gravitation causes dispersed matter to coalesce, and coalesced matter to remain intact, thus accounting for the existence of the earth, the sun, and most of the macroscopic objects in the universe.

Gravitation is responsible for keeping the earth and the other planets in their orbits around the sun; for keeping the moon in its orbit around the earth; for the formation of tides; for natural convection, by which fluid flow occurs under the influence of a density gradient and gravity; for heating the interiors of forming stars and planets to very high temperatures; and for various other phenomena observed on earth.

Gravitation is one of the four fundamental interactions of nature, along with electromagnetism, and the strong and weak forces. Modern physics describes gravitation using Einstein's general theory of relativity, in which gravitation is a consequence of the curvature of spacetime governing the motion of inertial objects. Newton's simpler law of universal gravitation provides an accurate approximation for most physical situations.

Gravity gradiometry is the study and measurement of variations in acceleration due to gravity. The gravity gradient is the spatial rate of change of gravitational acceleration.

Gravity gradiometry is used by oil, gas, and mining companies to measure the density of the subsurface, effectively the rate of change of rock properties. From this information it is possible to build a picture of subsurface anomalies which can then be used to more accurately target oil, gas, and mineral deposits. It is also used to image water column density when locating submerged objects, or determining water depth (bathymetry).

The unit of the gravity gradient is the Eotvos, which is equivalent to  $10^{-9} \text{ s}^{-2}$  (or  $10^{-4} \text{ mGal/m}$ ). A person walking past a point at a distance of 2 meters would provide a gravity gradient signal of approximately one Eotvos. Mountains can give signals of several hundred Eotvos.

## 4. Input data

The input data has two parts.

- (a). Input parameter file.
- (b). Input data file (includes migration domain file).

### (1). Input parameter file 'input\_gg\_mig.m'

This file includes the gravity and/or gravity gradiometry data indicator.

- (a). filename -- input file name,  
*i.e. filename = 'test2.dat';*

- (b). migid -- indication of the component(s) of the gravity field (mGal) and gravity gradiometry data (Eotvos) in the input data file (see part (4.2).input file), The indication is a number or a group of numbers. The meaning is:
  - 1 -- Gz -- the vertical component of the gravity field (mGal),
  - 2,3,4,5,6,7 -- Gxx, Gxy, Gxz, Gyy, Gyz, Gzz -- a component of gravity gradiometry data (Eotvos), respectively.

8 -- Gdelta =(Gxx-Gyy)/2

The order of the numbers corresponds to the order of the components in the input file (see part (4.2).input file).

*i.e. migid=[1, 7, 8];*

- (c). migjoint – indication of the joint migration components.

If *migjoint* is empty ( *migjoint* = [ ] ; ), the code will do migration for each single component that's indicated by *migid* (see (b) ), it would not do any joint migration;

If *migjoint* is not empty, the code will do joint migration as *migjoint* indicates.

*i.e. migjoint=[1, 7, 8];*

The number indicated by *migjoint* must be included in the *migid*, otherwise, it will show an error message.

- (d). modelid -- indication for migration domain information,  
 The indication can be 0 or 1.  
 0 -- means the migration domain will be defined by the following parameters  
 (see (e)~(k)).  
 1 -- means the x and y (center) coordinates of the migration domain will be  
 defined by an input file (see part (4.3) domainfile), the z coordinates will  
 are defined as (g)~(i).  
*i.e. modelid=0;*

The migration domain parameters if modelid=0:

- (e). xstep -- migration domain cell size in x direction,  
*i.e. xstep=100;*  
 (f). ystep -- migration domain cell size in y direction,  
*i.e. ystep=100;*

The minimum and maximum values will be set up by the receiver range.

- (g). nz -- mesh number of z direction,  
*i.e. nz = 20;*  
 (h). zmin -- The minimum value of z depth of the migration domain (m),  
*i.e. zmin = 50;*  
 (i). zmax -- The maximum value of z depth of the migration domain (m),  
*i.e. zmax = 1000;*  
 (j). delta\_x --- observed data interval in x direction (m),  
*i.e. delta\_x = 50;*  
 (k). delta\_y --- observed data interval in y direction (m),  
*i.e. delta\_y = 50;*

The migration domain parameters if modelid=1:

- (l). domainfile --- input file name for defining the migration domain (for format, see  
 part (4.3) domainfile),  
*i.e. domainfile = 'test2d.dat';*
- (m). restrictid --- indicator if needed to restrict the output density.  
 The indicator can be 0 or 1.  
 0 -- means no restriction to the output density,  
 1 -- means to restrict the output density data between ro\_min and ro\_max,  
*i.e. restrictid = 1;*
- (n). ro\_min --- minimum value to restrict the output density. Only needed when  
 restrictid = 1,  
*i.e. ro\_min = 0;*
- (o). ro\_max --- maximum value to restrict the output density. Only needed when  
 restrictid = 1,  
*i.e. ro\_max = 0.4;*

## (2). data input file

This file includes observed data coordinates and gravity and/or gravity gradiometry data values.

Input format: (5 numbers for each line at least).

x, y, z, topo, Gz, Gxx, Gzz,...

x --- x coordinate of receiver (m),

y --- y coordinate of receiver (m),

z --- z coordinate of receiver (m) (positive upwards),

clearance --- fly clearance of receiver to the ground surface (m) (always positive),

Gz --- vertical component of gravity field (mGal),

Gxx --- component of gravity gradient field (E),

Gzz --- component of gravity gradient field (E),

...

note: topography (elevation) (m) (positive upwards) = z – clearance.

### (3). migration domain input file

This data file only exists when modelid = 1.

The file contains x and y (center) coordinates for each column of migration cells.

Input format: (2 numbers for each line)

x --- x coordinate of center migration cell (m),

y --- y coordinate of center migration cell (m).

Note: The z coordinate information still needs to be input (see (g)~(i) ).

## 5. Output data

The code will write the density result to the file 'den\_'+filename or 'denj\_'+filename.

Output format: (4 numbers in each line at least)

x, y, z, den

x --- x coordinate of a model cell (m) (center),

y --- y coordinate of a model cell (m) (center),

z --- z coordinate of a model cell (m) (center),

den --- density in the cell.

If the *migjoint* is not empty, the den value is the joint migration density; output file is 'denj\_'+filename.

If the *migjoint* is empty, the den value is migration density corresponding to each component indicated in *migid* (see 4.1(b) ). The output file is 'den\_'+filename.

## 6. Operation

The code has two files: 'gg\_migration.m' and 'input\_gg\_mig.m'.

The file 'input\_gg\_mig.m' includes the input parameter, which needs to be created before running the migration (see the input data).

In the MATLAB environment (run 'matlab' to start the MATLAB environment).

Type 'gg\_migration'+ 'ENTER'.

The code will start running.

The code will open the 'input\_gg\_mig.m' file and input the data from the file which is indicated in the filename.

The code will show the process (percentage) and write the result to the file 'den\_'+filename or 'denj\_'+filename.

## 7. Example

There are two examples included in the package.

### Example 1.

A prism under a mountain.

'test1.dat' is the Full Tensor Gravity field data.

'den\_test1.dat' is the 3D density distribution of single component migration.

'denj\_test1.dat' is the 3D density distribution of joint migration.

'test1\_field.jpg' shows what the observed field looks like.

'test1.jpg' shows the center vertical cross section of the migration result.

### Example 2.

Two prisms under a valley.

'test2.dat' is the Full Tensor Gravity field data.

'den\_test2.dat' is the 3D density distribution of single component migration.

'denj\_test2.dat' is the 3D density distribution of joint migration.

'test2\_field.jpg' shows what the observed field looks like.

'test2.jpg' shows the center vertical cross section of the migration result.

## 8. References:

Zhdanov, M. S., Xiaojun Liu, Le Wan, and Glenn A. Wilson, 2011, Potential field migration for rapid imaging of gravity gradiometry data: Geophysical Prospecting, **59**, 1052–1071.

## 9. Contact information

In case of questions or suggestions, please feel free to contact the authors:

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